

CSC 226 Exercise 3
An Nguyen, V01073373
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Problem:

Prove the theorem: Let G be a connected edge-weighted undirected graph with distinct edge weights, and let C be some cycle in G . Then the maximum-weight edge e in C is not contained in the minimum weight spanning tree of G .

Proof:

We will prove the theorem by contradiction. We know that in a connected graph G , where all edge weights are distinct, there is a unique minimum weight spanning tree (MWST). If an edge satisfies to be the minimum weight bridge for a specific cut, it must be in that unique tree.

Claim:

Let C be a cycle in G , and let e be the maximum-weight edge in that cycle. We want to prove that e is not in the MWST.

Assumption:

Suppose, for the purpose of contradiction, e is in the MWST. Deleting any edge from a spanning tree disconnects it into exactly two connected components, which we can call S and $V - S$. By deleting e from our assumed tree, we create such a partition, or a cut, in the graph. Since e was in the cycle C , there must be another edge f in C that connects S and $V - S$ so that the cycle can cross the cut at e and go through the cut again at f .

From the assumption, we have that e is the maximum-weight edge in C . Since all edge weights in G are distinct, any other edge in C , including f , must have a weight less than that of e .

By the Cut Property, for any partition of a graph, the minimum weight edge crossing the cut must be in the MWST. In our case S and $V - S$ form a cut, and we have edge f crossing the cut with weight less than that of e . Therefore, edge f must be in the MWST. Since we also assumed that e is in the MWST, we have a contradiction: a spanning tree cannot contain a cycle.

Conclusion:

Thus, our assumption that e is in the MWST must be false. Therefore, the maximum-weight edge e in cycle C is not contained in the minimum weight spanning tree of G .